Intermediate (temperature and cost, side by side) — Solution

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Story — `temperature_cost_practice.md`: `temperature` controls how varied (random) the answers are — low for repeatable, high for creative — while the token count decides what each call costs. Seeing both side by side with real numbers is the base for every later cost-vs-quality decision. **If not:** you'd tune these by feel.

Solution

Story: the same two parts, now with the mechanism behind `temperature` (it reshapes the next-token probability distribution — flatter means more variety) and the units spelled out in the cost sum (price per 1K tokens × number of tokens). **If not:** a units mistake would put your cost estimate off by a factor of a million.

Part 1 — temperature, explained mechanically. At each step, the model computes a probability distribution over its whole vocabulary for "what comes next." `temperature=0` collapses that into effectively always taking the single highest-probability token — this is called greedy decoding, and it's why repeated calls converge on nearly identical output. `temperature=1` samples from the distribution closer to its natural shape, so tokens with real but lower probability get picked sometimes, and one different token early in the sentence compounds into a different sentence by the end, since each next token is chosen conditioned on everything before it.

This is also why a fact-lookup or classification prompt barely moves with temperature — the model's distribution is already sharply peaked on one answer regardless of setting, so there's nothing for temperature to meaningfully redistribute. An open-ended creative prompt has a much flatter distribution across many reasonable continuations, so temperature has real material to work with.

Part 2 — cost, with units made explicit.

```
input_rate_per_million  = $0.15
output_rate_per_million = $0.60

input_tokens  = 2000
output_tokens = 500

cost = (2000 / 1_000_000) * 0.15 + (500 / 1_000_000) * 0.60
```

```
      = 0.0003 + 0.0003
      = $0.0006
```

Scaling this up matters more than the tiny absolute number suggests: at 10,000 calls a day with this shape, that's $6/day, or about $180/month, from one feature alone — and that's before a growing conversation history multiplies the input-token count on every later turn.